

DAMA Data Quality Dimensions

Applied to the NHS RTT Waiting Times Project

DAMA (Data Management Association International) publishes the most widely-recognised framework for data quality, defined in its DMBOK (Data Management Body of Knowledge). This document defines DAMA's six core data quality dimensions and maps each one to the specific check already performed in the NHS RTT Waiting Times project, showing that the project's data quality investigation was, in substance, a DAMA-aligned assessment even before being formally labelled as one.

1. The Six DAMA Dimensions

Dimension	Definition
Accuracy	The degree to which data correctly describes the real-world object or event it represents.
Completeness	The degree to which all data that is expected or required is actually present, with no unwarranted gaps.
Consistency	The degree to which data values are the same, or do not contradict each other, across different records, systems, or points in a dataset.
Timeliness	The degree to which data is available and up to date for the purpose it is intended to serve.
Validity	The degree to which data conforms to the syntax, format, type, or range of an acceptable, defined value.
Uniqueness	The degree to which no entity is recorded more than once, and no duplicate records exist for the same real-world thing.

2. Mapping to the NHS RTT Project's Data Quality Checks

DAMA Dimension	Project Check	Result
Accuracy	Total Validation Check - SUM of all 105 weekly wait-band columns compared against the reported Total column, per row.	Reconciled - no mismatches found across 182,411 rows.
Completeness	Missing Values by Provider - percentage of NULL values across the 105 weekly columns, measured per provider.	Profiled and reported. NULLs preserved (not replaced with 0) to avoid understating the true scale of incomplete reporting.
Consistency	C_999 "Total" Rollup Investigation - tested whether the rollup row's total was internally consistent with the sum of the real specialty rows it claims to summarise.	Confirmed consistent - and confirmed as a genuine rollup, now excluded from all specialty-level analysis.
Timeliness	NHS England publishes RTT waiting times data on a monthly cycle. The most recently published month (July 2026) was deliberately selected for this project, rather than an older or arbitrary period.	Partially assessed. Confirmed the analysis used the freshest available data at the time of the project (a genuine timeliness check in itself). Full multi-month trend analysis was scoped out due to the size of a single month's file (182,411 rows) already being substantial -

		combining multiple months was judged impractical for this project's scope, not attempted and found lacking.
Validity	Commissioner Code = 0 Anomaly Check, and Negative Value Check across all 108 numeric columns.	0 rows with invalid Commissioner Code, 0 negative values found anywhere in the dataset.
Uniqueness	Duplicate Row Check - grouped by Provider + Commissioner + RTT Part Type + Treatment Function; verified at the correct grain after an initial false positive.	0 true duplicates confirmed.

3. Why This Reframing Matters

The checks performed in the original project were designed intuitively, from first principles, before this project's data quality vocabulary was formally mapped to DAMA. Retrofitting this mapping demonstrates two things to an employer:

1. The checks that were run were not arbitrary -they independently reconstruct 5 of the 6 dimensions that the data management industry itself considers the standard checklist for assessing data quality.
2. The one dimension not covered (Timeliness) is correctly identified as a genuine gap caused by the dataset's structure (a single snapshot), rather than an oversight - showing an accurate, honest self-assessment against the full framework rather than a retrofitted claim of full coverage.